

AI-Based Thermal Anomaly Detection for Hydrogen Storage and Distribution Safety Monitoring

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Summary

This project proposes a passive, camera-based monitoring system that uses machine learning to detect early-stage leaks in hydrogen storage and distribution infrastructure — pipes, tanks, and valve manifolds — from thermal imagery alone. A first neural network segments each thermal frame into installation components (pipe, tank, valve) versus surrounding background. A second stage analyzes temperature deviations from an adaptive, per-pixel baseline to flag anomalies indicative of a leak, using change-detection methods designed to catch slow, sustained drifts that simple thresholds miss. The system is designed to raise an alarm for human verification rather than to act autonomously, fitting naturally into existing industrial safety workflows.

Motivation

Hydrogen's role in renewable energy storage (produced from surplus renewable generation and reconverted to electricity on demand) is growing, but hydrogen's wide flammability range and small molecule size make leak detection at fittings and valves a persistent safety and reliability challenge. Hydrogen does not absorb well in the optical or infrared spectrum, so it cannot be imaged directly the way methane or SF₆ can with existing gas-imaging cameras. Most current detection relies on point sensors (no spatial coverage) or manual technician inspection. A passive thermal-imaging approach, informed by hydrogen's distinct thermal expansion behavior (it warms rather than cools on expansion at ambient temperature, unlike most other gases), offers a low-cost, scalable, and gas-informative alternative for continuous monitoring of critical infrastructure.

Proposed Approach

The system pairs low-cost thermal cameras with an embedded/edge computer running two coordinated models: a segmentation network isolating installation components from background, and an anomaly-detection network operating on adaptive-baseline temperature deviations within each region. Distinguishing background from component surfaces allows tighter anomaly thresholds where nothing should ever be warm or cold, improving sensitivity while controlling false alarms from legitimate operational and environmental temperature drift.

Intellectual Merit

The project advances learned spatiotemporal anomaly detection for cluttered, drifting industrial thermal environments, and investigates whether the sign and magnitude of a thermal deviation can help discriminate leak substance (e.g., hydrogen's atypical warming-on-expansion behavior versus the cooling behavior of most other process gases). This connects fundamental learning-system research to a concrete energy-infrastructure reliability problem.

Broader Impacts

Low-cost, passive safety monitoring for hydrogen and other energy infrastructure supports safer adoption of hydrogen-based renewable energy storage, and the sensing/edge-AI architecture is transferable to other industrial monitoring contexts.